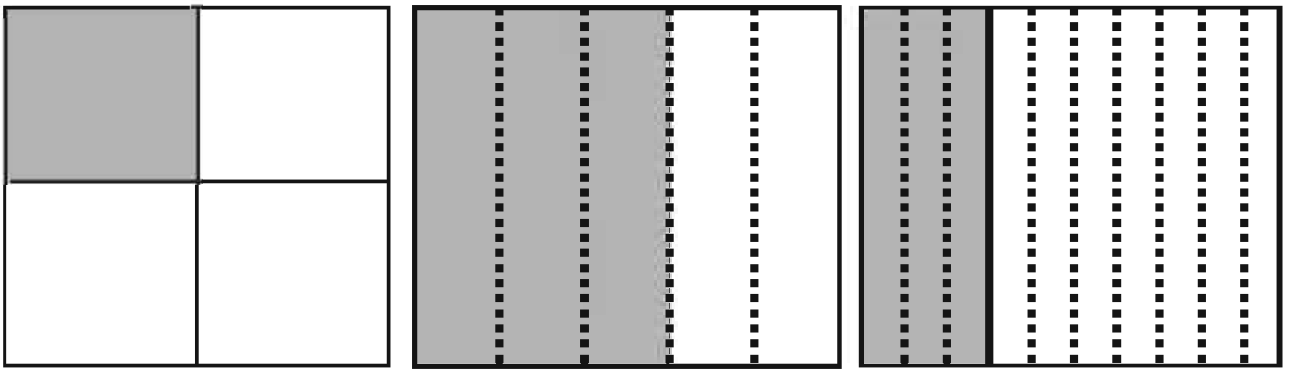
15. The ratio of milk and sugar in frumenty is 7 : 2. In the frumenty the quantity of sugar is 4 kg. what is the quantity of milk ?

16. The ratio of the prices of two computers is 5 : 6. If the price of the first computer is Tk. 25,000/-, what is the price of the second computer ? In case of price hike if the price of the first computer is increased by Tk. 5,000/- then what types of ratio of their prices will be?

2.3 Relation between Ratio and Percentage



1 : 4 a 3 : 5 b 3 : 10 c

Of the above figures, $\frac{1}{4}$ parts of a, $\frac{3}{5}$ parts of b, and $\frac{3}{10}$ parts of c have been shaded.

Here, we see that,

in figure (a), the ratio of the shaded part and the whole part is

$$1:4 = \frac{1}{4} = \frac{1 \times 25}{4 \times 25} = \frac{25}{100} = 25\%,$$

in figure (b), the ratio of the shaded part and the whole part is

$$3:5 = \frac{3}{5} = \frac{3 \times 20}{5 \times 20} = \frac{60}{100} = 60\%,$$

in figure (c), the ratio of the shaded part and the whole part is

$$3:10 = \frac{3}{10} = \frac{3 \times 10}{10 \times 10} = \frac{30}{100} = 30\%,$$

That is, 25%, 60%, 30% of the figures a, b, c are shaded respectively. It is seen that both percentage and ratio are fractions. But, in case of percentage, the denominator of the fraction is 100. In case of ratio numerator and the denominator may be of any natural number. If necessary, the percentage can be expressed into ratio and the ratio can be expressed into percentage. For example, the ratio of Tk. 7 and Tk. 10

$$= \frac{\text{Tk.}7}{\text{Tk.}10} = \frac{7}{10} = \frac{70}{100} = 70\%$$

Here, Tk. 7 is $\frac{7}{10}$ part of Tk.10 or $\frac{7}{10}$ times of Tk.10, which is equal to 70%.

On the other hand, percentage of 3 or 3% is $\frac{3}{100}$ or 3 : 100.

That is, a ratio can be expressed into percentage.

Activity :

1. Express the ratios 3 : 4 and 5 : 7 to percentage.
2. Express 5% and 12% to ratio.

Example 7. Express into ratios and decimals.

(a) 15% (b) 32% (c) 25% (d) 55% (e) $8\frac{1}{10}\%$

Solution : (a) $15\% = \frac{15}{100} = \frac{3}{20} = 3 : 20 = \cdot 15$

$\therefore 15\% = 3 : 20 = \cdot 15$

(b) $32\% = \frac{\overset{8}{\cancel{32}}}{\underset{25}{\cancel{100}}} = \frac{8}{25} = 8 : 25 = \cdot 32$

$\therefore 32\% = 8 : 25 = \cdot 32$

(c) $25\% = \frac{25}{100} = \frac{1}{4} = 1 : 4 = \cdot 25$

$\therefore 25\% = 1 : 4 = \cdot 25$

(d) $55\% = \frac{55}{100} = \frac{11}{20} = 11 : 20 = \cdot 55$

$\therefore\ 55\% = 11 : 20 = \cdot 55$

(e) $8\frac{1}{10}\% = \frac{81}{10}\% = \frac{81}{10} \times \frac{1}{100} = \frac{81}{1000} = 81 : 1000 = 0.081$

$\therefore\ 8\frac{1}{10}\% = 81 : 1000 = \cdot 081$

Example 8. Express the following fractions into percentage.

(a) $\frac{1}{4}$ (b) $\frac{3}{20}$ (c) $\frac{7}{15}$ (d) $\frac{4}{25}$ (e) $\frac{6}{13}$

Solution : (a) $\frac{1}{4} = \frac{1 \times 100}{4 \times 100} = \frac{25}{100} = 25\%$

(b) $\frac{3}{20} = \frac{3 \times 100}{20 \times 100} = \frac{15}{100} = 15\%$

(c) $\frac{7}{15} = \frac{7 \times 100}{15 \times 100} = \frac{140}{3} \times \frac{1}{100} = \frac{140}{3}\% = 46\frac{2}{3}\%$

(d) $\frac{4}{25} = \frac{4 \times 100}{25 \times 100} = \frac{16}{100} = 16\%$

(e) $\frac{3}{13} = \frac{3 \times 100}{13 \times 100} = \frac{300}{13} \times \frac{1}{100} = \frac{300}{13}\% = 23\frac{1}{13}\%$

Example 9. One quantity is 50% of another quantity. Determine the ratio of two quantities.

Solution : $50\% = \frac{50}{100}$ that is, if one quantity is 50, then the other will be 100.

$\therefore$ Ratio of 50 and 100 is $50 : 100 = 1 : 2$.

$\therefore$ Required ratio of two quantities $= 1 : 2$.

Example 10. The sum of two quantities is 240. If their ratio is 1 : 3, then find them. What is the percentage of the first quantity to the second ?

Solution : the Sum of two quantities = 240

Their ratio = 1 : 3

The sum of two quantities of the ratio = 1 + 3 = 4

$\therefore$ 1st quantity = $\frac{1}{4}$ times of 240

$= \overset{60}{\cancel{240}} \times \frac{1}{\cancel{4}} = 60$

$\therefore$ 2nd quantity = $\frac{3}{4}$ times of 240

$= \overset{60}{\cancel{240}} \times \frac{3}{\cancel{4}} = 180$

Again, ratio of two quantities is 1 : 3

$\therefore$ 1st quantity = $\frac{1}{3}$ of 2nd quantity = $\frac{1\times100}{3\times100} = \frac{100}{3}\% = 33\frac{1}{3}\%$

$\therefore$ 1st quantity is $33\frac{1}{3}\%$ of the second.

Example 11. Monira has obtained 80% marks in the annual examinations. If the total marks of the examinations are 800, how many marks has Monira obtained ?

Solution : The marks obtained by Monira = 80% of 800 = $\frac{80}{100} \times 800 = 640$

$\therefore$ the marks obtained by Monira 640

Example 12. 180 fazley mangoes were bought from a fruits shop. After two days 9 mangoes rotted. How many mangoes are quite good ?

Solution : The total number of mangoes which were bought = 180.
the number of rotten mangoes = 9.

$\therefore$ the number of good mangoes = 180 – 9 = 171.

The ratio of good mangoes and total mangoes is $\frac{171}{180} = \frac{19}{20}$

$\therefore$ The percentage of good mangoes is $\frac{19\times100}{20} = 95.$